



ESTATE MIND

Group 5 - 4 DS 5

Team



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Overview

- I. BO/DSP
- II. Matching Data to objectives
- III. Data Collection Methods
- IV. Data Sources & Justification
- V. Architecture
- VI. Data Understanding
- VII. EDA
- VIII. Mockups & Scenario
- IX. Conclusion



01. BO / DSO

BO (Business Objective)	DSO (Data Science Objectives)
BO1 — Improve Market Reliability Increase transparency and quality of real-estate listings.	DSO1: Automate multi-source data ingestion. DSO2: Structure and clean listings using NLP/LLMs. DSO3: Track, version, and orchestrate data pipelines.
BO2 — Understand Territorial Dynamics Identify spatio-temporal trends and emerging areas.	DSO1: Model market activity with time-series methods. DSO2: Perform spatial aggregation and mapping of listings. DSO3: Detect emerging zones and trigger alerts.
BO3 — Estimate Prices with Confidence Deliver accurate and explainable property valuations.	DSO1: Predict prices using supervised learning models. DSO2: Integrate temporal and territorial drivers. DSO3: Explain predictions using XAI techniques. DSO4: Detect under- and over-valued properties. DSO5: Expose valuations and explanations through APIs and visual analytics.
BO4 — Support Investment Decisions Optimize choices through personalized analytics.	DSO1: Build user profiles from preferences and behavior. DSO2: Simulate profitability and “what-if” scenarios. DSO3: Recommend properties and strategies via scoring models.
BO5 — Secure Legal Compliance Reduce regulatory risk and automate conformity checks.	DSO1: Model legal rules in a regulatory knowledge graph. DSO2: Continuously update regulatory data. DSO3: Compute an automated legal-risk score. DSO4: Detect non-compliance and issue alerts. DSO5: Orchestrate specialized agents for continuous monitoring.
BO6 — Make the Platform Operational Enable access through AI assistants and automated reporting.	DSO1: Allow interaction via a conversational assistant. DSO2: Automatically generate reports and summaries. DSO3: Ensure explainability and traceability of outputs.

02. Matching Data to objectives

Matching Data ↔ BOs

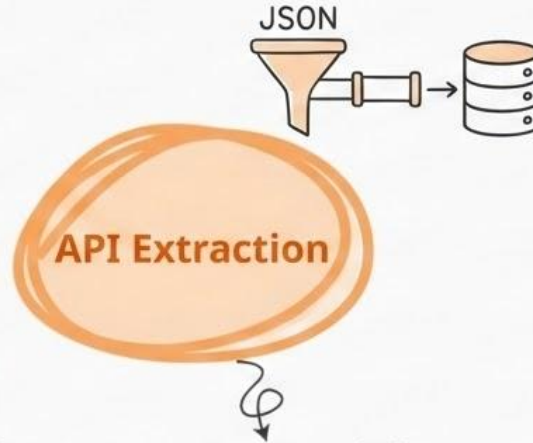
Business Objective	Key Data Needed	Purpose	Sources
BO1 — Market Reliability	IDs, timestamps	Ensure data quality & detect anomalies	Tayara, Mubawab...
BO2 — Territorial Dynamics	Location, coordinates, time, POIs	Analyze spatial & temporal trends	open data
BO3 — Price Estimation	Price, surface, rooms, location	Build predictive models	Tayara, Mubawab...
BO4 — Investment Support	Price, rental data, user preferences, economic data	Simulate ROI & recommend properties	open data
BO5 — Legal Compliance	Regulations, zoning rules, descriptions	Detect risks & non-compliance	banks , simulated data
BO6 — Platform Operation	All processed data, models, logs	Enable AI assistant & reporting	

03. Data Collection Methods

Collection Methods

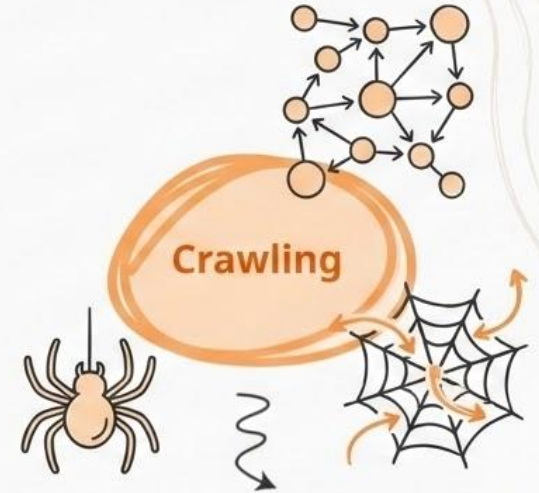


Most Tunisian platforms do not provide open APIs, so scraping is the most effective solution.



To retrieve structured data more efficiently.

For example, Tayara uses a Next.js architecture, which exposes JSON endpoints that we can directly query.



This ensures scalability while avoiding excessive requests.

04. Data Sources & Justification

Data sources

Source	Type	Data Type	Value	Limitations
Tayara	Classified platform	Semi-structured	High volume, recent data	Noisy, duplicates
Mubowab	Real estate platform	Semi-structured	Better structure, features	Medium volume
Remax	Agency	Structured	Reliable data	Low volume
Tecnocasa	Agency	Semi-structured	Detailed descriptions	Limited coverage
Century21	Agency	Structured	Professional data	Limited volume
TPS Immobilière	Local agency	Semi-structured	Local insights	Heterogeneous

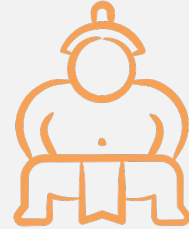
Justification



Reliability



Diversity

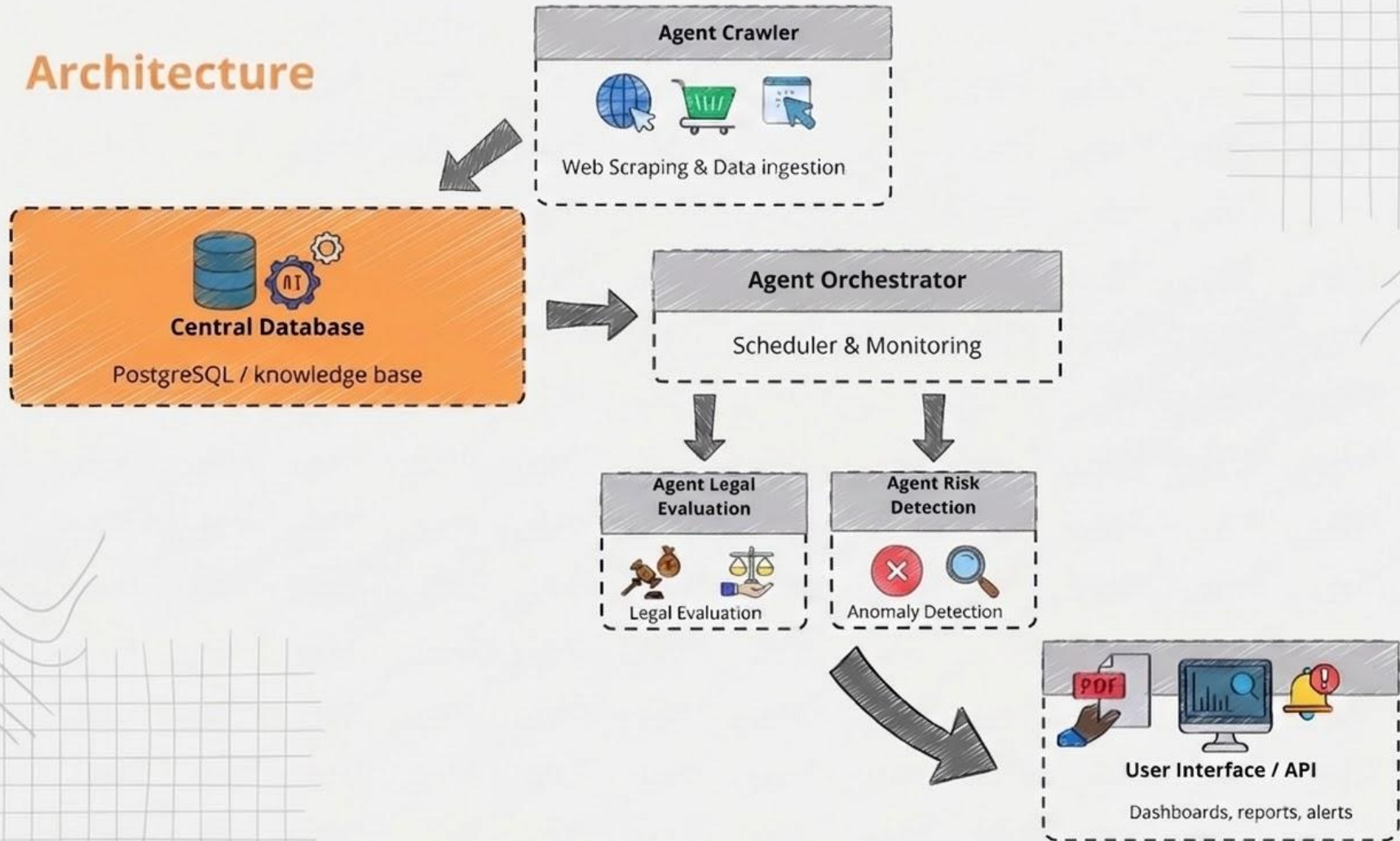


Volume

Platforms like Tayara and Mubawab provide large-scale and up-to-date data, while real-estate agencies such as Remax or Tecnocasa provide more reliable and structured information.

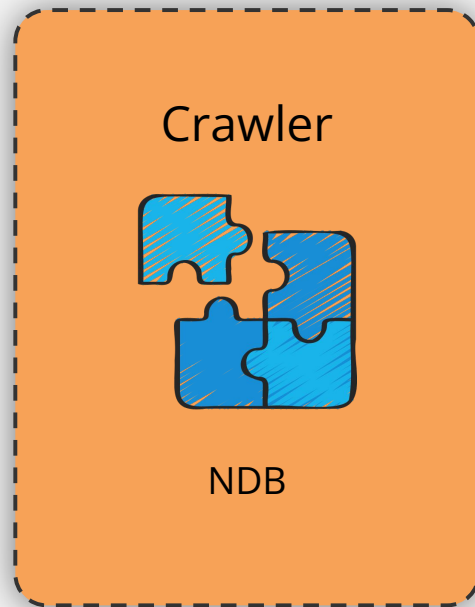
05. Architecture

Architecture



Agent Collector

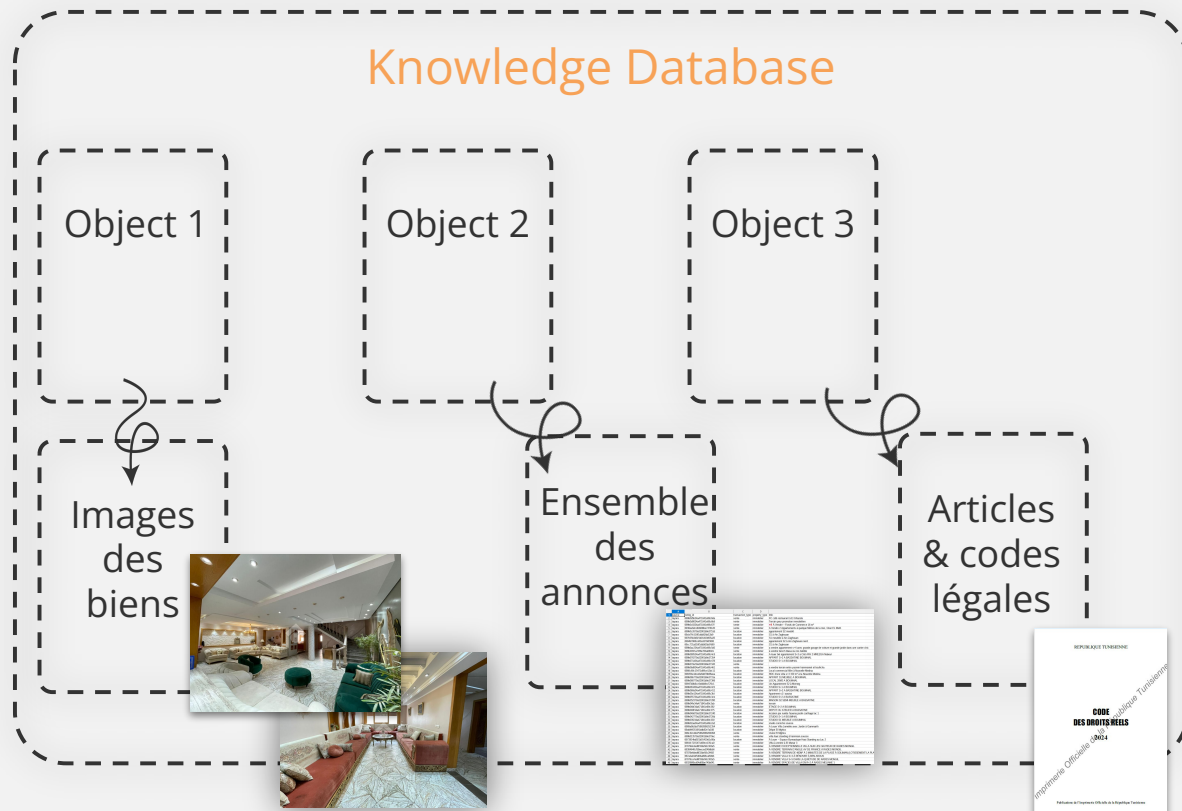
Sources	Data Collection Method
Tayara 	JSON data endpoint scraping (Next.js API)
Mubaweb 	HTML DOM scraping + detail page crawling
Tecnocasa 	REST API harvesting + HTML enrichment + detail page crawling
Remax 	REST Search API querying with pagination



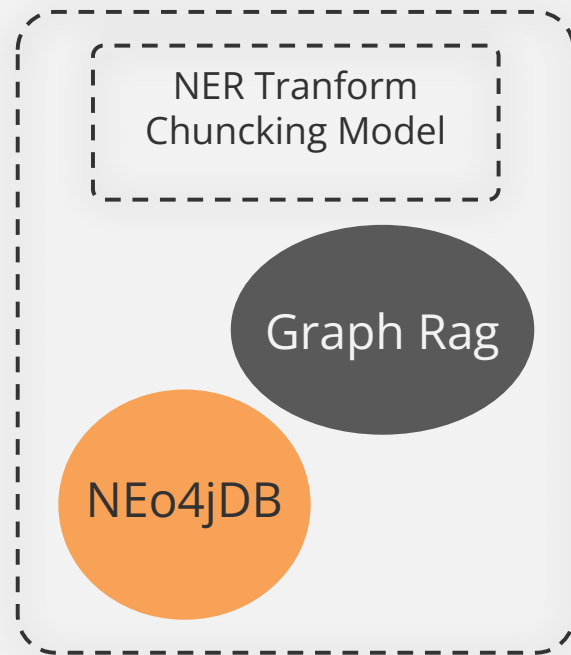
Knowledge Base

Agent Collector

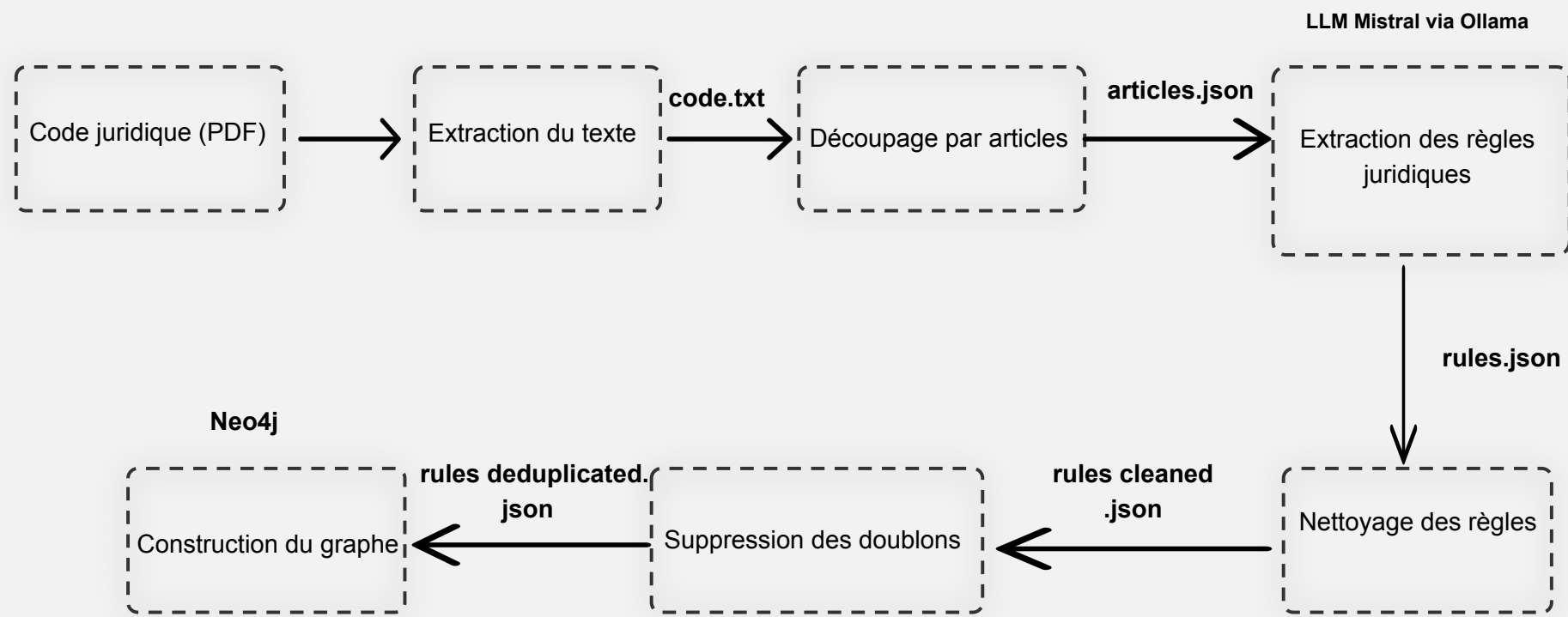
Knowledge Database



Agent Comply

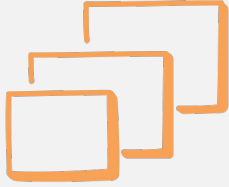


Architecture collecte des données juridique



06. Data Understanding

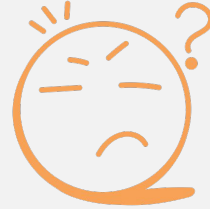
Cleaning & Standardisation



Duplicated
values



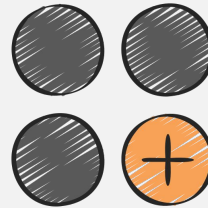
Missing
values



Outliers

01
02
03

Encoding



Standardisation

Quality Validation



Timeless

We ensure data is up to date by tracking scraping dates



Comprehensibility

We analyze the coverage of regions and property types



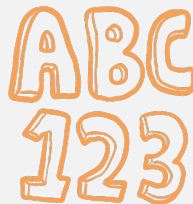
Relevance

We select only features useful for our project objectives



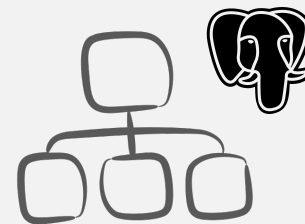
Validity

We apply validation rules to ensure data correctness



Consistency

We ensure consistent formats across all datasets

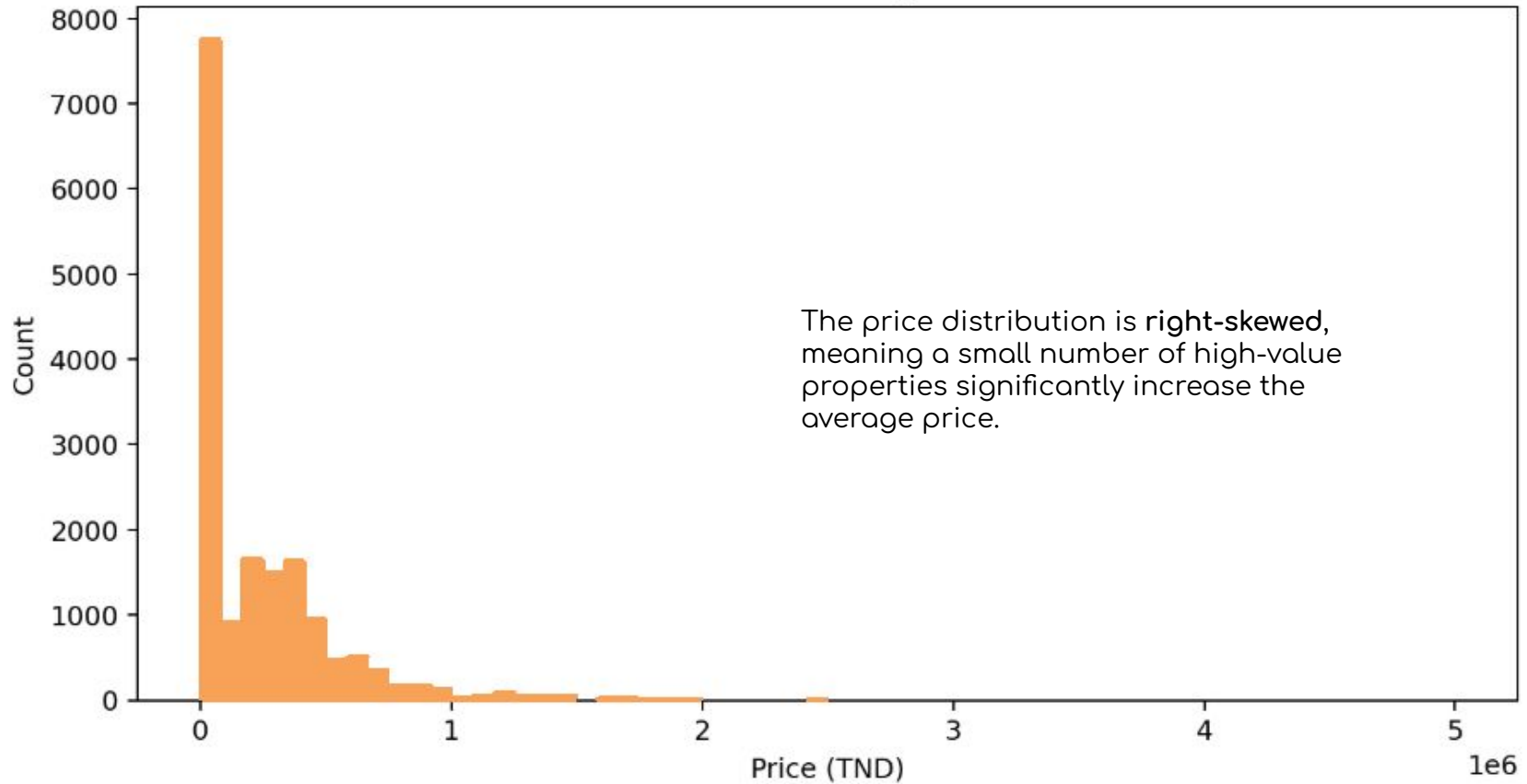


Accessibility

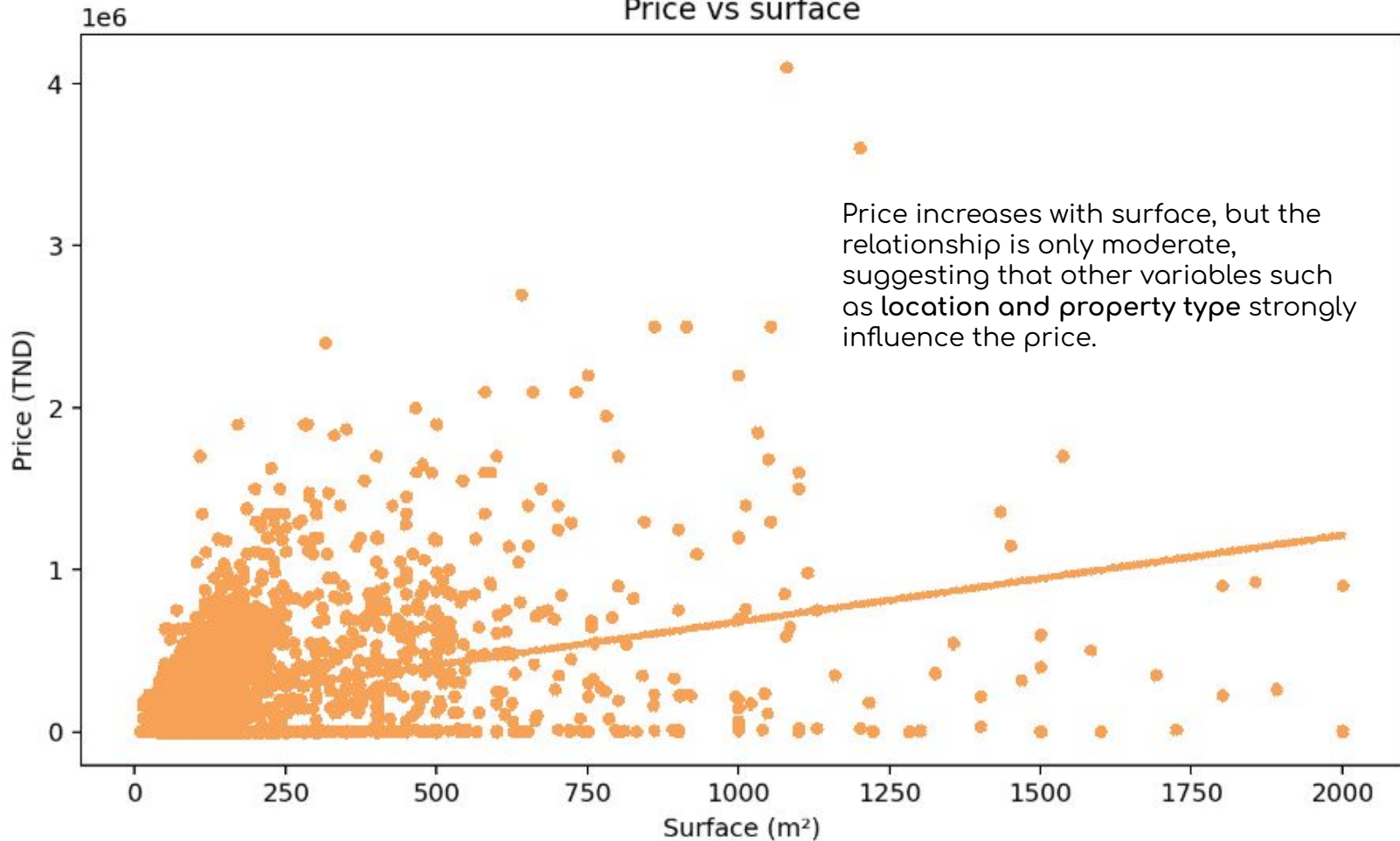
Data is stored in a structured database for easy access and reproducibility

07. EDA

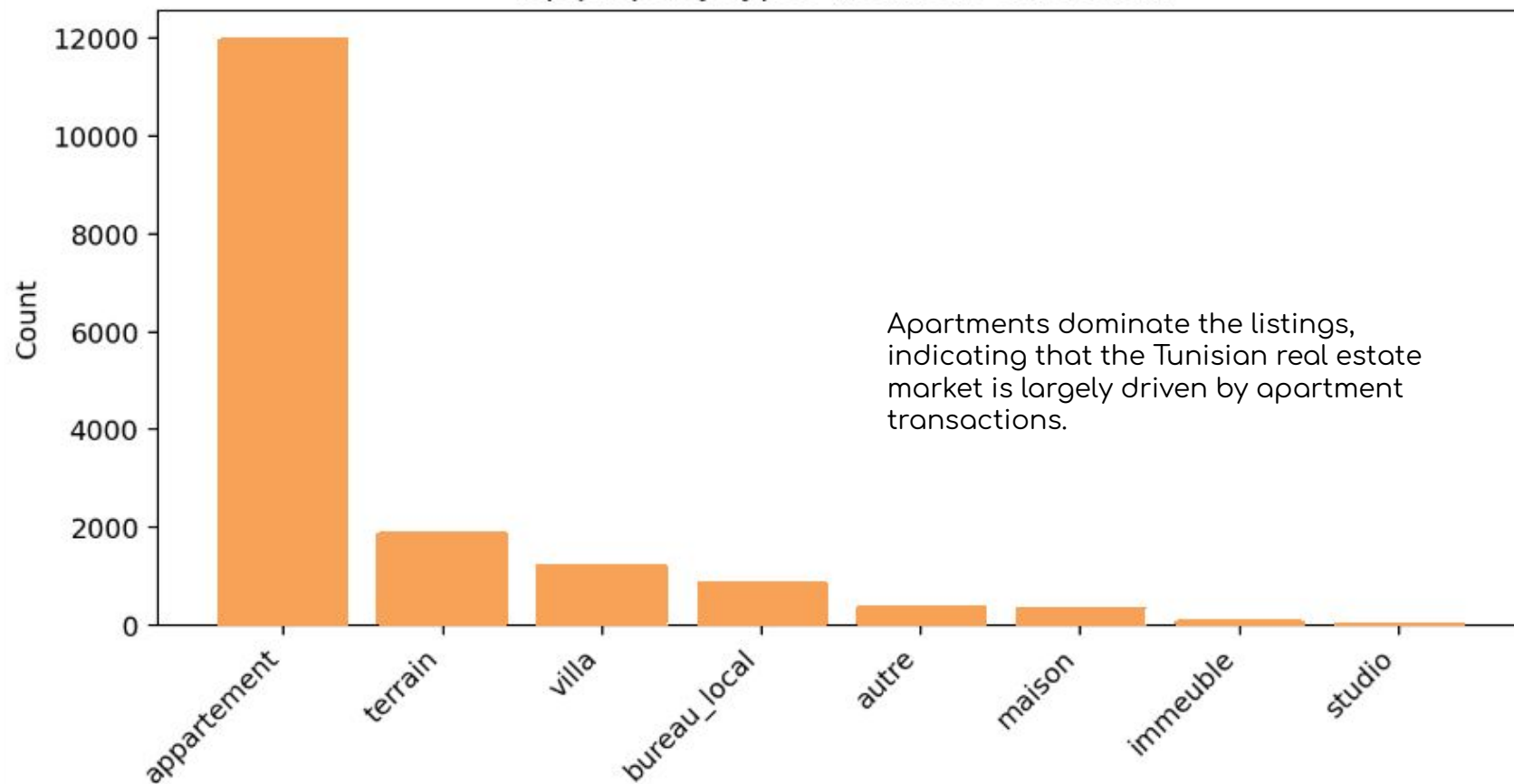
Distribution of prices



Price vs surface

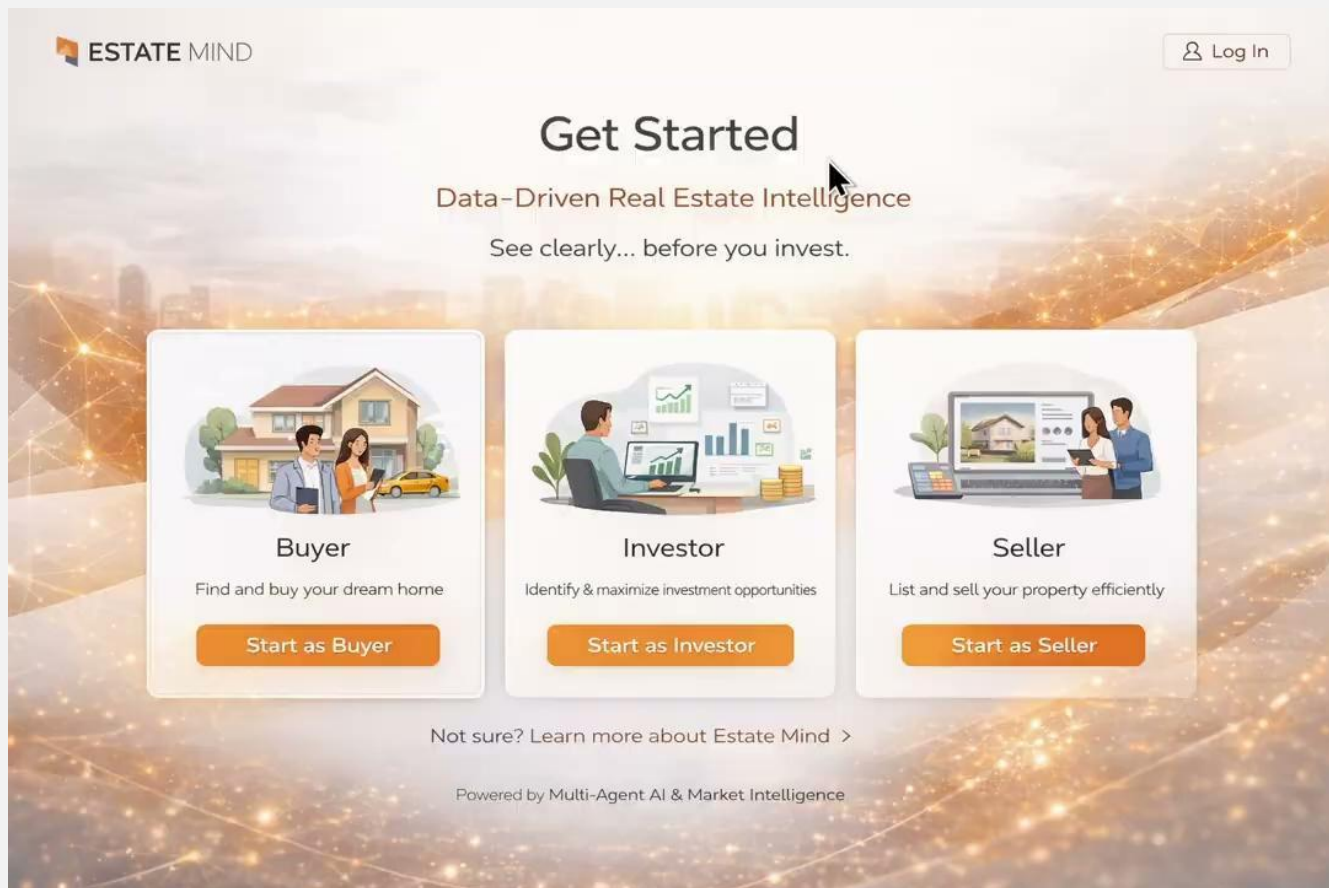


Top property types (heuristic extraction)



08. Mockups & Scenario

Mockups



https://youtu.be/rmSdukLx_z0

9. Conclusion

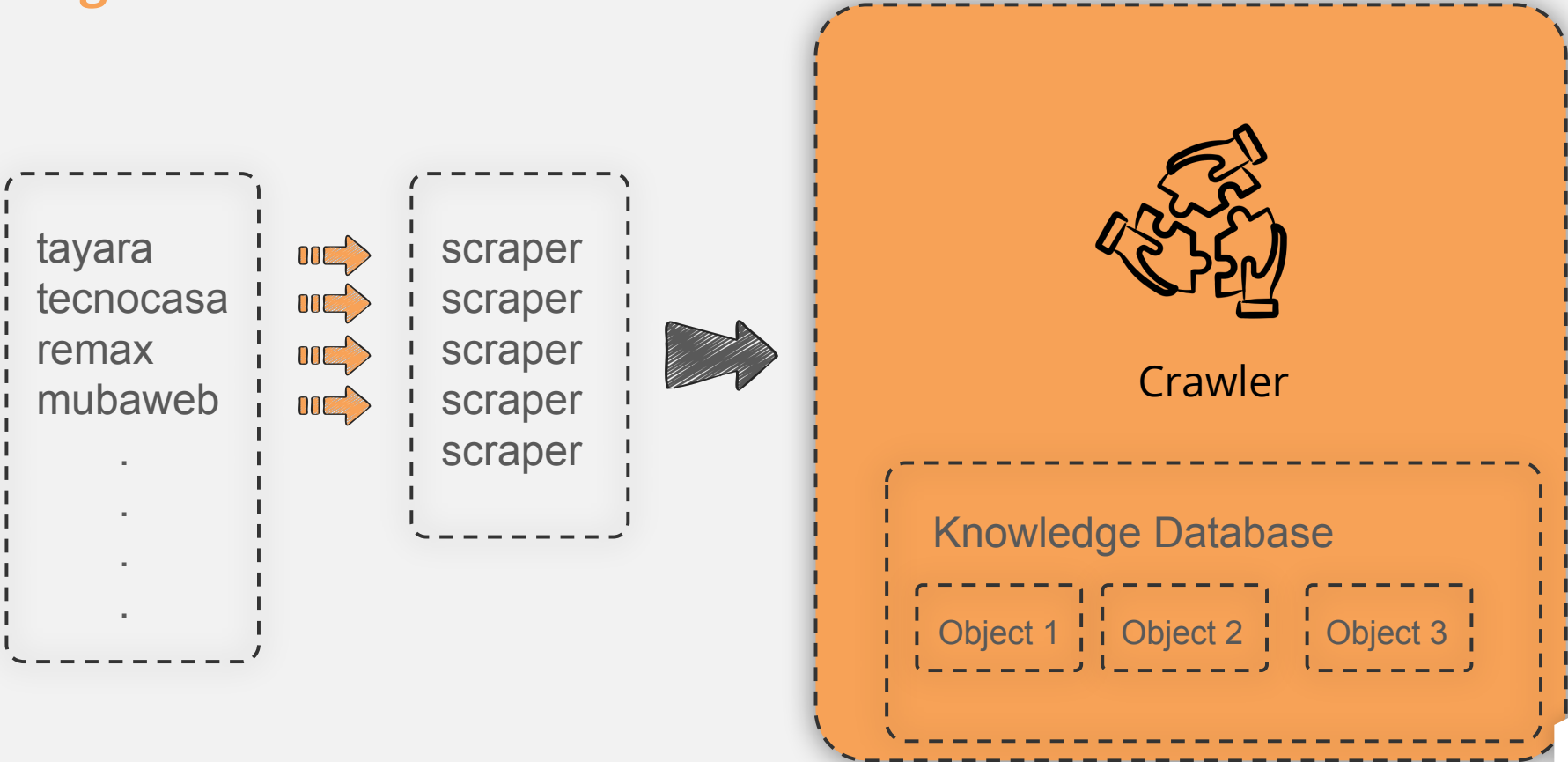


TuniEstate : See clearly...
before you invest.

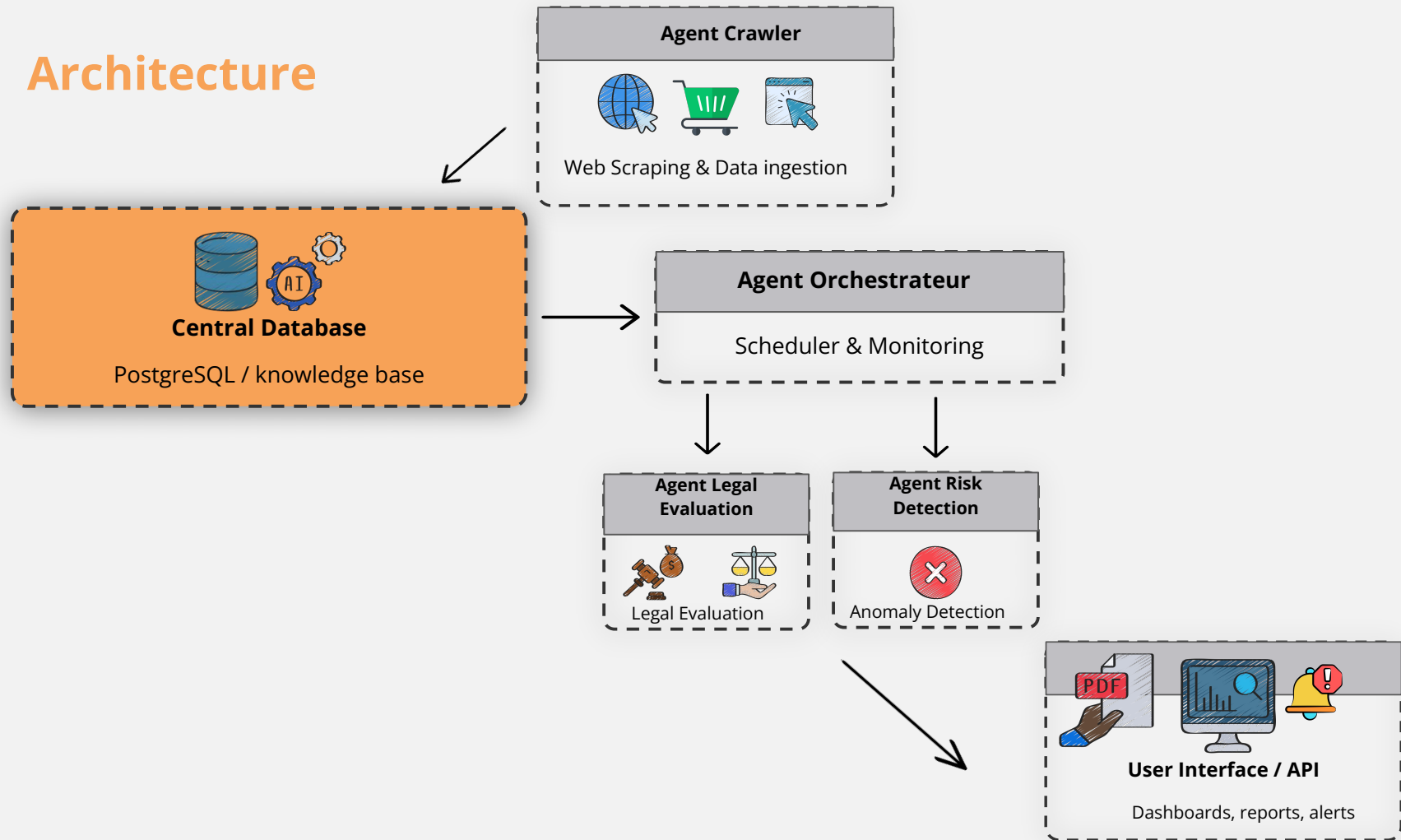
Thanks for your
attention !



Agent Collector



Architecture



Collection Methods

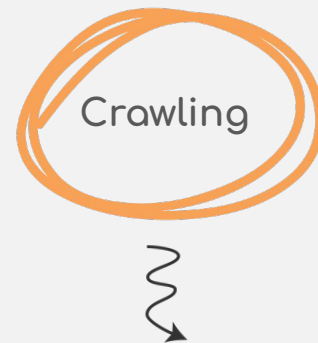


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